

Detailed Methodology

Sign language is not the same as natural language. It has its own grammar, and syntax. There are not any advanced systems which can translate natural language to sign language glosses (written format of sign language) efficiently. To solve this problem I used Supervised Fine tuning technique and fine-tuned 2 state of the art models to translate English text to American Sign Language glosses and made a rule based algorithm.

The 3 approaches are as follows:

1. Finetuning MarianMT
2. Finetuning T5
3. Rule based Algorithm

I used ASLG_PC12 dataset which have 87.7 K samples of text and gloss in ASL.

1. Finetuning Marian Mt

Firstly I chose the pre-trained model of Marian Mt I need to fine-tune. The language I found to be similar to English and glosses were German. So I took Marian MT opus-mt-en-de (English to German) model. Then the dataset is split into test, train and validation.

Then using Transformers *AutoTokenizer*, I tokenised all the dataset.

Then using ***AutoModelForSeq2SeqLM*** and ***Seq2SeqTrainer***, with learning rate $2e-5$ and running on 3 epochs, I finetuned the Marian Mt Model.

2. Finetuning T5 Model

Firstly the dataset is split into test, train and validation. Then using Transformers ***AutoTokenizer*** the dataset is tokenised. Then taking ***T5-Base*** model and ***Seq2SeqTrainer*** with learning rate $4e-5$, 2 epochs, 100 eval steps, 100 logging steps, and 200 save step, I started fine-tuning the base model.

3. Rule based Algorithm

Taking an input sentence, we clean that first, removing punctuations, converting to lower case, etc. Then the sentence is split into words. Now special functions check for temporal words, nouns, verb, adjectives and adverbs, and finally, WH Family words. Then all these words are combined in the following format

[temporal word, noun, verb, adjective/adverb, wh family]

Hence the sentence gets converted into a gloss. Note: Glosses removes some elements such as Articles, "ed" at last of words and in gloss all words are written in CAPITALS.

"What is your Name ?"  YOUR NAME WHAT
TEXT GLOSS